

Linear Algebra

Friday 2 October 2026 | Optional supplement | Student copy - no solutions

Only two optional tasks S1-S2. Attempt them only if your tutor selects them and time remains.

S1 Find, correct and check

Tutor-written extension. Try for 2 minutes; 5 minutes including discussion.

A student makes the row operation shown below. Find and correct the error, solve the original system, and verify your solution in both original equations.

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & -1 & 0 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -3 & 0 \end{array} \right]$$

Correction, solution and substitution check:

S2 Build your own examples

Tutor-written extension. Try for 3 minutes; 6 minutes including discussion.

Construct three systems, each with exactly three equations in two variables: (i) no solution; (ii) a unique solution; (iii) infinitely many solutions. Write their augmented matrices and justify each answer. Include a zero row in at least one consistent example.

Work over the real numbers. Three augmented matrices and reasons:

These two extensions add up to 11 minutes if both are used. They are optional, not part of the 50-minute main route.

